

Business Proposal

Many organizations collect large amounts of data but lack the tools necessary to transform that information into actionable insights. The Division of Purchasing and Supply (DPS) is one such organization, employing more than 100 staff members who manage procurement operations across the Commonwealth. Manually analyzing procurement data is challenging due to the volume of records and the need to evaluate spending across multiple time periods. Our team proposes a Business Intelligence (BI) solution that will establish a centralized system to store procurement records and provide users with a dashboard and predictive analysis functionalities. The system will enable DPS to identify spending trends more effectively while improving reporting accuracy and strategic decision-making.

The BI solution will benefit multiple stakeholders, with primary value delivered to DPS procurement analysts, executive leadership, and VADOC decision-makers. DPS analysts have the opportunity to use a central database that is continuously updated with all procurement data. The system will enable users to track expenditures, generate financial reports, and communicate insights effectively to leadership. Automating data preparation will allow analysts to focus more on generating business insights and identifying operational improvement opportunities.

The dashboards that will be created for DPS leaders and decision-makers will provide clear views of trends and future spending patterns, helping support better budgeting, cost management, and strategic planning. VADOC stakeholders and other affected groups will also gain greater visibility into procurement activity, supporting daily operations and resource planning. Overall, the BI solution will enhance efficiency, improve transparency, and promote more strategic use of data across DPS and VADOC.

Our solution is the most appropriate approach for us because it is concentrated on generating trustworthy, well-organized, and actionable insights from the complicated dataset, while remaining feasible as a student project for a semester. The system will be built using industry-standard BI tools and established data management practices, ensuring realistic and scalable performance. The sprint-based approach will allow us to evaluate our concepts and refine our models through data acquisition and stakeholder feedback.

The proposed BI solution may encounter resistance due to changes in work processes, hesitation toward new technology, and concerns about data accuracy. Employees may prefer to work with their current manual reports instead of using the new self-service dashboard system. Some stakeholders may question the accuracy of transformed data or predictive analytics, especially for high-visibility compliance reports such as SWAM spend. Security controls and access permissions of the centralized repository system will raise concerns among users. Users believe that they will face extra work from learning the new visualization tool instead of achieving improved efficiency. If not addressed early, these concerns could slow adoption and delay project progress.

The project will succeed through an agile, collaborative approach aligned with the sprint-based structure outlined in the RFP. The DPS stakeholders who matter most will assist us in selecting tools and designing dashboards to create a solution that meets actual business demands. The first sprints will deliver high-value outputs which show immediate benefits through total spend reports and compliance dashboards. We will provide structured training sessions and clear documentation to support user adoption. Transparent data validation processes will further build trust in the centralized repository. The combination of stakeholder engagement with incremental value creation and effective communication will result in better product adoption together with sustained operational success.

The BI solution will be easy for the customer to use because Tableau offers an intuitive, user-friendly interface that will require minimal technical expertise. Most of the complex setup will occur during development, so the customer will use the dashboards through simple filters and visuals after they refresh the data. Users can accomplish their tasks because the system provides basic functions which do not require them to possess technical expertise. The training session will provide sufficient information to help them navigate the dashboards and understand the essential insights. Continued system usage will be supported by the platform's intuitive design and familiar operating structure.

A detailed financial analysis is outside the scope of this proposal; however, the solution is relatively inexpensive because it leverages cloud-based BI tools and existing data infrastructure, which helps to reduce hardware expenses and enables the use of scalable licensing models. The main costs of the project will come from data cleaning operations and integration work as well as ETL automation implementation and user training activities that support change management. The system will maintain cost efficiency through automated processes that decrease the need for manual work, which applies to dashboard development, cloud storage expansion, and ongoing system upkeep.

The team will conduct a formal tool evaluation using a Pugh decision matrix to compare Tableau and Power BI across key decision criteria, including cost, scalability, security, ease of integration, user experience, and learning curve. The weighted scoring system will then indicate that Tableau achieves the highest overall score due to its superior visualization capabilities, interactive executive storytelling features, ability to manage large and complex datasets, and seamless data source integration that aligns with DPS/DOC procurement and compliance reporting requirements. Ultimately, our team has selected Tableau as the primary BI tool because its advanced analytics capabilities, intuitive design, and long-term strategic value will justify the higher licensing cost.

Epic Story #1 – Clarity in Reporting and Use of Data

As DPS analysts and members of executive leadership at VADOC, we want all procurement data to be stored in one centralized source and updated regularly so we can easily analyze spending, create reports, and reduce the time spent manually working with data.

Having this centralized system will allow us to track trends, improve reporting accuracy, meet compliance requirements (such as SWAM), and provide a clear and reliable view of procurement activity.

Epic Story #2 – Smarter Decision-Making and Strategic Planning

As DPS executives and stakeholders, we want access to easy-to-use dashboards and predictive tools that show patterns and trends in procurement data so we can make better budgeting, cost management, compliance, and spending decisions moving forward.

These tools will allow leadership to quickly understand current spending, identify unusual changes, and plan for future needs instead of only reviewing past data.

Our team has created two epic stories to guide the development of the BI solution and explain the main value the system will provide. The first epic focuses on improving reporting and making procurement data easier to use. DPS analysts and executive leadership need all procurement data from across the state to be stored in one central place that is reliable and regularly updated. This will allow them to analyze spending, create accurate reports, and spend less time working with data manually. Having one central source of information will also help support compliance needs such as SWAM reporting and give a clear view of procurement activity across the organization.

The second epic focuses on helping decision-makers make better decisions and plan for the future. DPS executives and stakeholders need simple dashboards and predictive tools that show trends and patterns in procurement data. These tools will help leaders understand how money is being spent now, notice any unusual changes, and predict future needs. With a clearer view of procurement data, decision-makers will be able to plan budgets better, manage costs, stay compliant with regulations, and make smarter long-term decisions instead of only relying on past reports.